

WASP-52b

R-0422

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_171221 · created 2026-07-19T08:56:21+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458108.6361 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.216 +/- 0.063
Transit depth (Rp/Rs)^2	4.65 +/- 2.7 [%]
Orbital Inclination (inc)	87.97 +/- 2.77
Airmass coefficient 1 (a1)	0.88 +/- 0.46
Airmass coefficient 2 (a2)	0.05 +/- 0.28
Scatter in the residuals of the lightcurve fit is	51.12 %
Best Comparison Star	None
Optimal Aperture	2.41
Optimal Annulus	4.25
Transit Duration (day)	0.085 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.216 ± 0.063 vs 0.1646 (deviation 0.82σ)
Duration fit vs published	0.085 d vs 0.0758 d (deviation 0.76σ)
Mid-time O–C	-8.8 min
Depth SNR	3.4
Residual scatter	51.12 %

Light curve

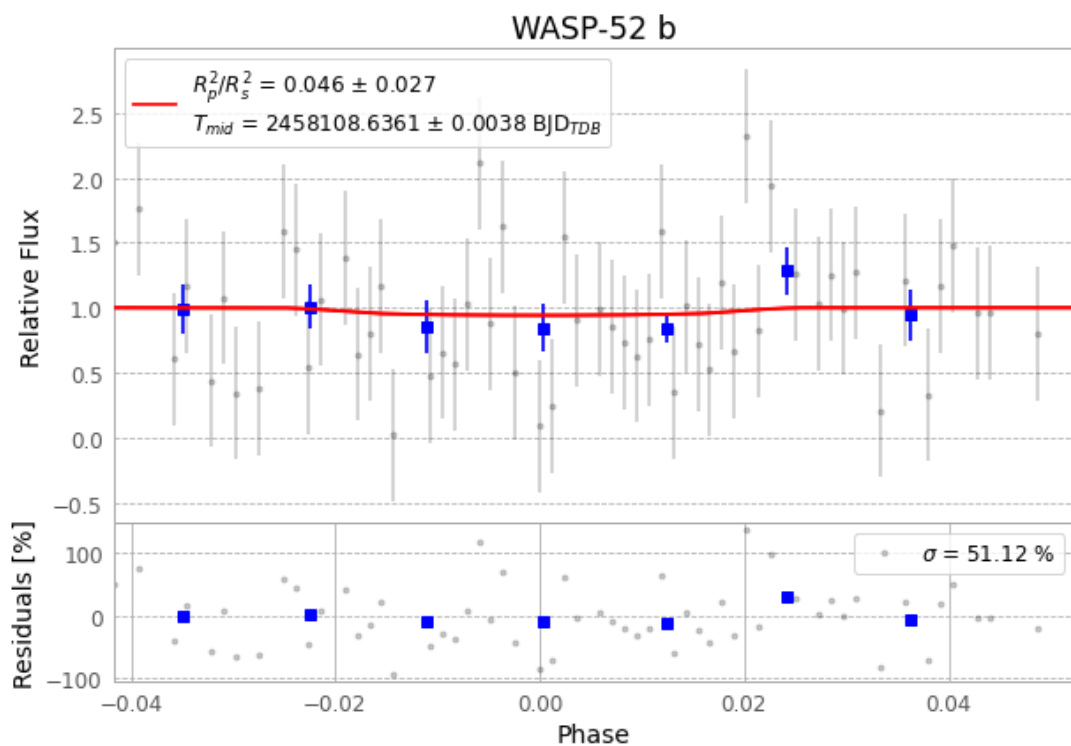


Figure 1 — FinalLightCurve_WASP-52 b_2017-12-20.png (SHA-256 f78d63ee58b7988e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [578, 109], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 d3e879e91e9cb361...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

80 FITS frames (53 MB), WASP-52171221013012.FITS → WASP-52171221052712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-171221.log (b8419375bb2c...), FinalLightCurve_WASP-52 b_2017-12-20.png (f78d63ee58b7...), FinalLightCurve_WASP-52 b_2017-12-20.csv (ababceed02d3...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 08:56Z.